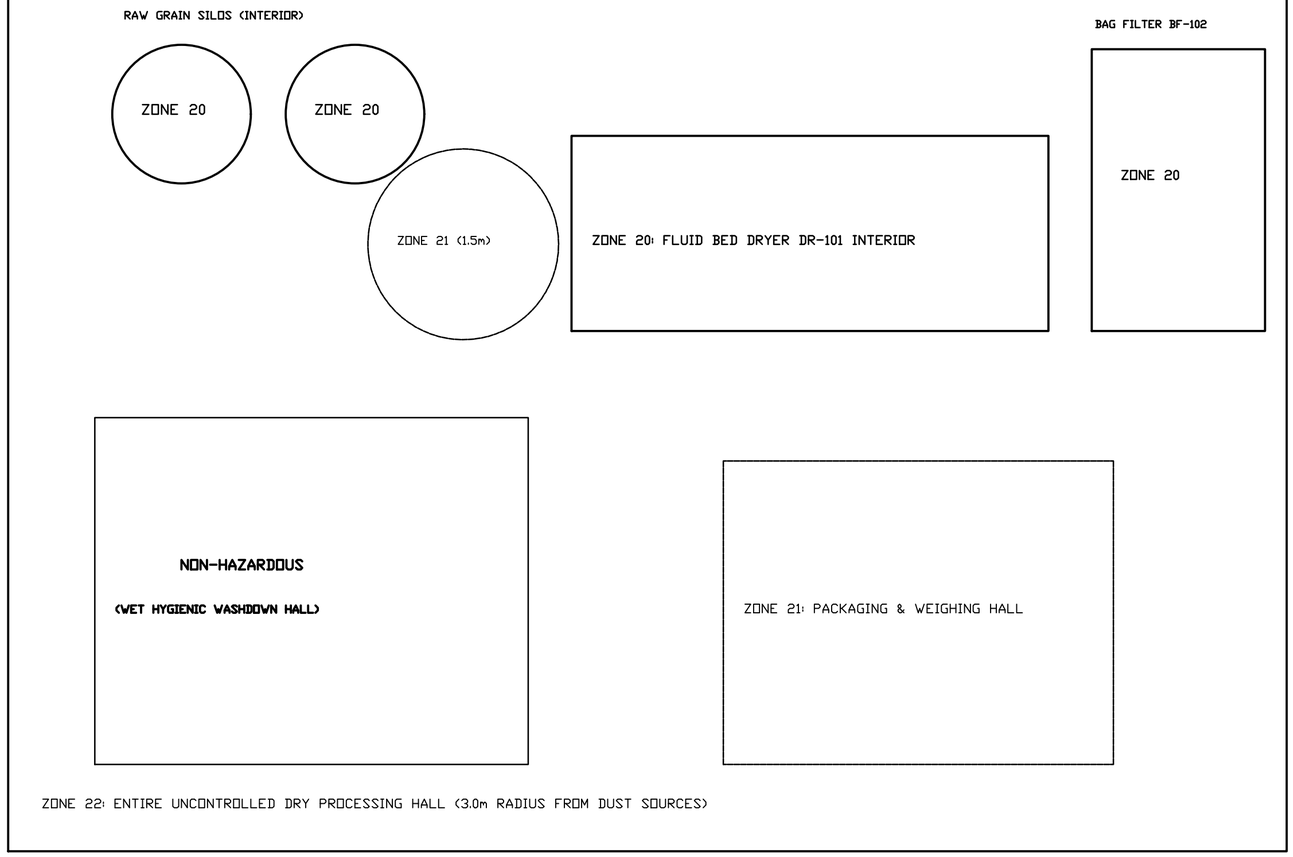
PROCESSING BUILDING - PLAN VIEW COMBUSTIBLE DUST ZONING (NFPA 652 / IEC 60079-10-2)



COMBUSTIBLE DUST CHARACTERISTICS (NFPA 652):

1. Material: Tanzanian Toor Dal Grain Dust (Cajanus cajan)
2. Dust Deflagration Class: St 1 (Kst: 85 bar-m/s, moderate)
3. Maximum Explosion Pressure Pmax: 7.6 bar g
4. Minimum Ignition Energy (MIE): > 30 mJ (No spark hazard)
5. Minimum Ignition Temp (MIT - Cloud): 430 deg C
6. Layer Ignition Temperature (LIT - 5mm): 260 deg C
7. Permissible Equipment Surface Temp: T120 deg C
   (Max 2/3 of MIT / 75K below LIT per IEC 60079-0)
8. Zone 20 Enclosures: IP65 Ex tb IIIC T120 deg C Da
9. Zone 21/22 Enclosures: IP55 Ex tc IIIC T120 deg C Dc
10. Explosion Relief: Dual certified NFPA 68 burst panels
    on DR-101 hood and flameless vents on BF-102.
11. Static Bonding: All pipes, ductwork, filter cages and
    vessels bonded to earthing grid (< 10 ohms resistance).
12. Housekeeping Standard: Scheduled vacuum cleaning
    preventing dust layer accumulation > 1.0 mm.

HAZARDOUS AREA ZONE CLASSIFICATION DEFINITIONS & APPLICABLE STANDARDS:

ZONE 20: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently (> 1,000 h/year). Equipment must be Category 1D / EPL Da certified.

ZONE 21: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur in normal operation occasionally (10 to 1,000 h/year). Equipment must be Category 2D / EPL Db certified.

ZONE 22: A place in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but, if it does occur, will persist for a short period only (< 10 h/year). Category 3D / EPL Dc.

REGULATORY COMPLIANCE: NFPA 652 (Standard on the Fundamentals of Combustible Dust), NFPA 68 (Standard on Explosion Protection by Deflagration Venting), IEC 60079-10-2 (Explosive Atmospheres - Dust Area Classification), OSHA 1910.307.

FINNOAQ AGRO-INDUSTRIAL TECHNOLOGIES LTD.
15,000 kg/day Split Pigeon Pea Optimization Greenfield Project

HAZARDOUS AREA CLASSIFICATION & DUST ZONING
DISCIPLINE: SAFETY | LEAD: SOURISH SENAPATI | GUIDANCE: DR. ARCHISMAN RAY (IIT KGP)

DWG NO: DWG-FIN-15T-012	REV: V8.0	SHEET: 12 OF 13
SCALE: AS SHOWN DATE: 2026-10-17	STATUS: PRELIMINARY	GATE: G-A HOLD

CONCEPTUAL FEED - PRELIMINARY - NOT FOR CONSTRUCTION / NOT FOR PROCUREMENT | COPYRIGHT FINNOAQ